DIVIDE AND CONQUER

1-Number of Zeros in a Given Array

```
#include <stdio.h>
int main()
{
    int n, a[100], count = 0;
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for (int i = 0; i < n; i++)
        if (a[i] == 0)
            count++;
    printf("%d", count);
    return 0;
}
```

2-Majority Element

```
#include <stdio.h>
int main()
{
    int n, a[100], t;
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for (int i = 0; i < n - 1; i++)
```

```
        for (int j = i + 1; j < n; j++)
            if (a[i] > a[j])
            {
                t = a[i];
                a[i] = a[j];
                a[j] = t;
            }
    printf("%d\n", a[n / 2]);
    return 0;
}
```

3-Finding Floor Value

```
#include <stdio.h>
int main()
{
    int n, a[100], x, floor = -1;
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for (int i = 0; i < n; i++)
    {
        if (a[i] <= x)
            floor = a[i];
        else
            break;
    }
```

```
    printf("%d\n", floor);
    return 0;
}
```

4-Two Elements sum to x

```
#include <stdio.h>
int main()
{
    int n, a[100], x;
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &x);
    for (int i = 0; i < n; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (a[i] + a[j] == x)
            {
                printf("%d\n%d\n", a[i], a[j]);
                return 0;
            }
        }
    }
    printf("No\n");
    return 0;
}
```

5-Implementation of Quick Sort

```
#include <stdio.h>
void quickSort(int a[], int n)
{
    for (int i = 0; i < n - 1; i++)
    {
        for (int j = 0; j < n - i - 1; j++)
        {
            if (a[j] > a[j + 1])
            {
                int t = a[j];
                a[j] = a[j + 1];
                a[j + 1] = t;
            }
        }
    }
}
int main()
{
    int n, a[100];
    scanf("%d", &n);
    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    quickSort(a, n);
    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
```

```
}
```